

Machine Learning Prediction Assignment Writeup

R Markdown

Introduction The goal of this project is to predict the manner in which participants performed barbell lifts, represented by the “classe” variable in the training dataset. We’ll use data from accelerometers on the belt, forearm, arm, and dumbbell of 6 participants to build our prediction model.:

```
# Load required libraries
library(caret)

## Loading required package: ggplot2
## Loading required package: lattice
library(randomForest)

## randomForest 4.6-14
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
## The following object is masked from 'package:ggplot2':
##
##      margin
# Load the data
training <- read.csv("https://d396qusza40orc.cloudfront.net/predmachlearn/pml-training.csv")
testing <- read.csv("https://d396qusza40orc.cloudfront.net/predmachlearn/pml-testing.csv")

# Remove near zero variance predictors
nzv <- nearZeroVar(training)
training <- training[, -nzv]
testing <- testing[, -nzv]

# Remove columns with more than 95% NA values
na_cols <- sapply(training, function(x) mean(is.na(x))) > 0.95
training <- training[, !na_cols]
testing <- testing[, !na_cols]

# Remove irrelevant columns (first 7 columns)
training <- training[, -(1:7)]
testing <- testing[, -(1:7)]

# Ensure 'classe' is a factor
training$classe <- as.factor(training$classe)
```

Model Building

We'll use Random Forest for our prediction model due to its high accuracy and ability to handle complex interactions between features.

```
# Set seed for reproducibility
set.seed(123)

# Split training data into training and validation sets
inTrain <- createDataPartition(training$classe, p = 0.7, list = FALSE)
train_set <- training[inTrain, ]
val_set <- training[-inTrain, ]

# Train Random Forest model
rf_model <- randomForest(classe ~ ., data = train_set, ntree = 500)

#MODEL EVALUATION

# Make predictions on validation set
val_pred <- predict(rf_model, val_set)

# Calculate accuracy
accuracy <- confusionMatrix(val_pred, val_set$classe)$overall['Accuracy']
print(paste("Validation Accuracy:", round(accuracy, 4)))

## [1] "Validation Accuracy: 0.9947"

# Calculate out-of-sample error estimate
oos_error <- 1 - accuracy
print(paste("Estimated Out-of-Sample Error:", round(oos_error, 4)))

## [1] "Estimated Out-of-Sample Error: 0.0053"

#Predicting 20 test cases

# Make predictions on the 20 test cases
test_pred <- predict(rf_model, testing)
#print(test_pred)
```